

Course Catalogue

Complete Course & Lesson Plan

39	7	32	405
Total Courses	Complete	Pending	Total Lessons

Table of Contents

Programming Languages

- Java Mastery — 5 lessons · 40h
- Python for Everyone — 5 lessons · 28h
- Modern JavaScript — 5 lessons · 32h
- TypeScript Deep Dive — 5 lessons · 24h
- Go Essentials — 5 lessons · 30h
- Rust Systems Programming — 5 lessons · 50h

Frontend Development

- React 19 & Ecosystem — 5 lessons · 36h
- Next.js App Router — 5 lessons · 30h
- Angular 18 Enterprise — 5 lessons · 42h
- Vue.js 3 Composition API — 5 lessons · 28h
- Tailwind CSS Mastery — 5 lessons · 16h

Backend Development

- Node.js & Express — 5 lessons · 32h
- Spring Boot 3 — 5 lessons · 44h
- Django & DRF — 5 lessons · 34h
- FastAPI & Async Python — 5 lessons · 24h
- Microservices Architecture — 5 lessons · 40h

Cloud Computing

- AWS Solutions Architect — 5 lessons · 50h

DevOps & Infrastructure

- Docker & Containerization — 5 lessons · 18h
- Kubernetes in Production — 5 lessons · 48h
- Terraform & IaC — 5 lessons · 28h

AI & Machine Learning

- Python for AI & ML — 35 lessons · 45h
- PyTorch Deep Learning — 35 lessons · 55h
- TensorFlow & Keras — 35 lessons · 50h
- Hugging Face Transformers — 35 lessons · 45h
- Large Language Models — 35 lessons · 60h
- Retrieval-Augmented Generation — 35 lessons · 40h
- AI Agents & Agentic Workflows — 35 lessons · 50h
- Machine Learning Fundamentals — 5 lessons · 38h
- Generative AI & LLMs — 5 lessons · 30h
- RAG Systems & Vector Search — 5 lessons · 24h
- AI Agents & Tool Use — 5 lessons · 28h
- Advanced Prompt Engineering — 5 lessons · 12h
- MLOps & Model Deployment — 5 lessons · 36h

Databases

- PostgreSQL Advanced — 5 lessons · 28h
- MongoDB & Atlas — 5 lessons · 22h

Software Engineering

- Data Structures & Algorithms — 5 lessons · 60h

■ System Design — 5 lessons · 40h

Mobile Development

■ React Native — 5 lessons · 36h

■ Flutter & Dart — 5 lessons · 40h

Programming Languages

Java Mastery

■ Pending

Slug: `java` · Difficulty: **Intermediate** · Duration: **40h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Java, OOP, JVM, Concurrency, Streams, Java 21

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Python for Everyone

■ Pending

Slug: `python` · Difficulty: **Beginner** · Duration: **28h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Python, Scripting, Automation, Data

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Modern JavaScript

■ Pending

Slug: `javascript` · Difficulty: **Intermediate** · Duration: **32h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: ES2024, Async, DOM, Modules, Closures

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

TypeScript Deep Dive

■ Pending

Slug: `typescript` · Difficulty: **Intermediate** · Duration: **24h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Types, Generics, Decorators, Utility Types

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP

Go Essentials

■ Pending

Slug: `go` · Difficulty: **Intermediate** · Duration: **30h planned · 3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Goroutines, Channels, Interfaces, Microservices

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Rust Systems Programming

■ Pending

Slug: `rust` · Difficulty: **Advanced** · Duration: **50h planned · 3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Ownership, Borrow Checker, Concurrency, WASM

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Frontend Development

React 19 & Ecosystem

■ Pending

Slug: **react** · Difficulty: **Intermediate** · Duration: **36h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Hooks, Context, Suspense, RSC, TanStack

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Next.js App Router

■ Pending

Slug: **nextjs** · Difficulty: **Intermediate** · Duration: **30h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: App Router, RSC, ISR, Server Actions

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Angular 18 Enterprise

■ Pending

Slug: **angular** · Difficulty: **Intermediate** · Duration: **42h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Signals, Standalone, RxJS, NgRx

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Vue.js 3 Composition API

■ Pending

Slug: **vuejs** · Difficulty: **Beginner** · Duration: **28h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Composition API, Pinia, Nuxt 3, TypeScript

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP

Tailwind CSS Mastery

■ Pending

Slug: `tailwindcss` · Difficulty: **Beginner** · Duration: 16h planned · 3h lessons · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Utility-First, Design Systems, Animations

Module 1			
01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Backend Development

Node.js & Express

■ Pending

Slug: **nodejs** · Difficulty: **Intermediate** · Duration: **32h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: REST, Express, Fastify, Event Loop, Streams

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Spring Boot 3

■ Pending

Slug: **spring-boot** · Difficulty: **Intermediate** · Duration: **44h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Spring MVC, Security, JPA, Microservices, Reactive

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Django & DRF

■ Pending

Slug: **django** · Difficulty: **Intermediate** · Duration: **34h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: DRF, Authentication, ORM, Celery, Channels

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

FastAPI & Async Python

■ Pending

Slug: **fastapi** · Difficulty: **Intermediate** · Duration: **24h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Async, Pydantic v2, OpenAPI, Dependency Injection

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP

Microservices Architecture

■ Pending

Slug: **microservices** · Difficulty: **Advanced** · Duration: **40h planned** · **3h lessons** · Lessons: **5** (■2 ■1 ■1 ■1) · Tags: **DDD**, **Event-Driven**, **Saga**, **CQRS**, **Service Mesh**

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Cloud Computing

AWS Solutions Architect

■ Pending

Slug: **aws** · Difficulty: **Intermediate** · Duration: **50h planned** · **3h lessons** · Lessons: **5** (■2 ■1 ■1 ■1) · Tags: EC2, S3, Lambda, RDS, VPC, CDK

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

DevOps & Infrastructure

Docker & Containerization

■ Pending

Slug: **docker** · Difficulty: **Beginner** · Duration: **18h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Dockerfile, Compose, Networking, Security

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Kubernetes in Production

■ Pending

Slug: **kubernetes** · Difficulty: **Advanced** · Duration: **48h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Pods, Helm, Operators, RBAC, GitOps, Istio

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Terraform & IaC

■ Pending

Slug: **terraform** · Difficulty: **Intermediate** · Duration: **28h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: HCL, Modules, Remote State, Multi-Cloud

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

AI & Machine Learning

Python for AI & ML

■ Complete

Slug: `python-for-ai-ml` · Difficulty: **Beginner** · Duration: 45h planned · 22h lessons · Lessons: 35 (■29 ■4 ■0 ■2) · Tags: Python, NumPy, Pandas, Scikit-Learn, Deep Learning

Module 1

01	Introduction to Python and AI/ML Fundamentals	■ Reading	30 min · 75 XP
02	Python Basics: Variables, Data Types, and Operations	■ Reading	30 min · 75 XP
03	Control Flow: Conditionals and Loops	■ Reading	30 min · 75 XP
04	Functions and Modules in Python	■ Reading	30 min · 75 XP
05	Working with Lists, Tuples, and Dictionaries	■ Reading	30 min · 75 XP

Module 2

06	File Handling and Input/Output Operations	■ Reading	30 min · 75 XP
07	Object-Oriented Programming Fundamentals	■ Reading	35 min · 75 XP
08	Exception Handling and Debugging	■ Reading	30 min · 75 XP
09	NumPy: Creating and Manipulating Arrays	■ Reading	35 min · 75 XP
10	Foundation Concepts Review and Practice	■ Exercise	50 min · 100 XP

Module 3

11	Pandas: Data Structures and Loading Data	■ Reading	35 min · 75 XP
12	Exploratory Data Analysis with Pandas	■ Reading	35 min · 75 XP
13	Data Cleaning and Preprocessing Techniques	■ Reading	35 min · 75 XP
14	Feature Engineering and Selection	■ Reading	35 min · 75 XP
15	Statistical Analysis and Hypothesis Testing	■ Reading	35 min · 75 XP

Module 4

16	Data Visualization with Matplotlib and Seaborn	■ Reading	35 min · 75 XP
17	Introduction to Scikit-Learn and Model Building	■ Reading	35 min · 75 XP
18	Core Skills Assessment and Practice	■ Exercise	50 min · 100 XP
19	Supervised Learning: Regression Models	■ Reading	35 min · 75 XP
20	End-to-End Machine Learning Project: House Price Prediction	■ Project	60 min · 150 XP

Module 5

21	Supervised Learning: Classification Models	■ Reading	35 min · 75 XP
22	Model Evaluation Metrics and Cross-Validation	■ Reading	35 min · 75 XP
23	Hyperparameter Tuning and Grid Search	■ Reading	35 min · 75 XP
24	Ensemble Methods: Bagging and Boosting	■ Reading	35 min · 75 XP
25	Intermediate Machine Learning Practice	■ Exercise	50 min · 100 XP

Module 6

26	Unsupervised Learning: Clustering Algorithms	■ Reading	35 min · 75 XP
27	Dimensionality Reduction Techniques	■ Reading	35 min · 75 XP
28	Introduction to Neural Networks and Deep Learning	■ Reading	35 min · 75 XP
29	Building Neural Networks with TensorFlow and Keras	■ Reading	35 min · 75 XP
30	Advanced Topics Assessment and Practice	■ Exercise	50 min · 100 XP

Module 7

31	Convolutional Neural Networks for Image Processing	■ Reading	35 min · 75 XP
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32	Recurrent Neural Networks and Sequence Modeling	■ Reading	35 min · 75 XP
33	Natural Language Processing with Python	■ Reading	35 min · 75 XP
34	Model Deployment and Production Considerations	■ Reading	35 min · 75 XP
35	Comprehensive AI/ML Capstone Project	■ Project	60 min · 150 XP

PyTorch Deep Learning

■ Complete

Slug: **pytorch-deep-learning** · Difficulty: **Intermediate** · Duration: **55h planned** · **22h lessons** · Lessons: **35** (■29 ■4 ■0 ■2) · Tags: PyTorch, CNNs, RNNs, GANs, Deployment

Module 1

01	Introduction to PyTorch and the Deep Learning Ecosystem	■ Reading	30 min · 75 XP
02	Tensors: Creation, Operations, and GPU Acceleration	■ Reading	35 min · 75 XP
03	Autograd and Automatic Differentiation	■ Reading	35 min · 75 XP
04	Building Neural Networks with nn.Module	■ Reading	35 min · 75 XP
05	Loss Functions and Optimizers	■ Reading	30 min · 75 XP

Module 2

06	Training and Validation Loops	■ Reading	35 min · 75 XP
07	Data Loading with Dataset and DataLoader	■ Reading	35 min · 75 XP
08	Transfer Learning with Pretrained Models	■ Reading	35 min · 75 XP
09	Convolutional Neural Networks (CNNs)	■ Reading	35 min · 75 XP
10	Foundational Skills Exercise	■ Exercise	50 min · 100 XP

Module 3

11	Recurrent Neural Networks and LSTMs	■ Reading	35 min · 75 XP
12	Attention Mechanisms and Transformers in PyTorch	■ Reading	35 min · 75 XP
13	Custom Loss Functions and Metrics	■ Reading	35 min · 75 XP
14	Regularization: Dropout, Batch Norm, Weight Decay	■ Reading	35 min · 75 XP
15	Learning Rate Scheduling and Warm-Up Strategies	■ Reading	35 min · 75 XP

Module 4

16	Mixed Precision Training with torch.cuda.amp	■ Reading	35 min · 75 XP
17	Distributed Training with DistributedDataParallel	■ Reading	35 min · 75 XP
18	Intermediate Skills Checkpoint	■ Exercise	50 min · 100 XP
19	PyTorch Lightning for Clean Training Code	■ Reading	35 min · 75 XP
20	End-to-End Project: Image Classification Pipeline	■ Project	60 min · 150 XP

Module 5

21	Object Detection with PyTorch	■ Reading	35 min · 75 XP
22	Semantic Segmentation Models	■ Reading	35 min · 75 XP
23	Time Series Forecasting with PyTorch	■ Reading	35 min · 75 XP
24	Generative Adversarial Networks (GANs)	■ Reading	35 min · 75 XP
25	Advanced Techniques Exercise	■ Exercise	50 min · 100 XP

Module 6

26	Variational Autoencoders (VAEs)	■ Reading	35 min · 75 XP
27	Model Pruning and Quantization	■ Reading	35 min · 75 XP
28	TorchScript and Model Export	■ Reading	35 min · 75 XP
29	ONNX Export and Cross-Platform Deployment	■ Reading	35 min · 75 XP

30	Optimization Exercise: Latency and Throughput	■ Exercise	50 min · 100 XP
Module 7			
31	PyTorch Mobile and Edge Deployment	■ Reading	35 min · 75 XP
32	Custom CUDA Extensions	■ Reading	35 min · 75 XP
33	Debugging and Profiling PyTorch Models	■ Reading	35 min · 75 XP
34	Testing Neural Networks Effectively	■ Reading	35 min · 75 XP
35	Capstone: Production Deep Learning System	■ Project	60 min · 150 XP

TensorFlow & Keras

■ Complete

Slug: [tensorflow-keras](#) · Difficulty: **Intermediate** · Duration: **50h planned · 21h lessons** · Lessons: **35** (■29 ■4 ■0 ■2) ·
 Tags: TensorFlow, Keras, CNNs, Transfer Learning, TFLite

Module 1			
01	Introduction to TensorFlow and Keras Ecosystem	■ Reading	30 min · 75 XP
02	Understanding Tensors and Data Structures	■ Reading	30 min · 75 XP
03	Setting Up Your TensorFlow Development Environment	■ Reading	20 min · 50 XP
04	NumPy Fundamentals for TensorFlow	■ Reading	30 min · 75 XP
05	Computational Graphs and Static vs Dynamic Execution	■ Reading	35 min · 75 XP
Module 2			
06	Keras Sequential API Basics	■ Reading	30 min · 75 XP
07	Dense Layers and Activation Functions	■ Reading	30 min · 75 XP
08	Loss Functions and Optimization Algorithms	■ Reading	35 min · 75 XP
09	Training, Validation, and Test Sets	■ Reading	30 min · 75 XP
10	Foundation Concepts Review and Practice	■ Exercise	50 min · 100 XP
Module 3			
11	Convolutional Neural Networks Architecture	■ Reading	35 min · 75 XP
12	Pooling and Flattening Operations	■ Reading	30 min · 75 XP
13	Regularization Techniques: Dropout and L1/L2	■ Reading	35 min · 75 XP
14	Batch Normalization and Layer Normalization	■ Reading	35 min · 75 XP
15	Recurrent Neural Networks and LSTMs	■ Reading	35 min · 75 XP
Module 4			
16	GRU Cells and Bidirectional RNNs	■ Reading	35 min · 75 XP
17	Transfer Learning with Pretrained Models	■ Reading	35 min · 75 XP
18	Core Skills Application Exercise	■ Exercise	50 min · 100 XP
19	Building Custom Layers and Models	■ Reading	35 min · 75 XP
20	Image Classification Project: CIFAR-10 Dataset	■ Project	60 min · 150 XP
Module 5			
21	Data Augmentation and Preprocessing Pipelines	■ Reading	35 min · 75 XP
22	Hyperparameter Tuning and Grid Search	■ Reading	35 min · 75 XP
23	Early Stopping and Learning Rate Scheduling	■ Reading	35 min · 75 XP
24	Model Evaluation Metrics and Confusion Matrices	■ Reading	35 min · 75 XP
25	Core Skills Mastery Exercise	■ Exercise	50 min · 100 XP
Module 6			
26	Functional API for Complex Model Architectures	■ Reading	35 min · 75 XP

27	Multi-Input and Multi-Output Models	■ Reading	35 min · 75 XP
28	Attention Mechanisms and Transformers	■ Reading	35 min · 75 XP
29	Natural Language Processing with Keras	■ Reading	35 min · 75 XP
30	Advanced Techniques Application Exercise	■ Exercise	50 min · 100 XP
Module 7			
31	Model Deployment and TensorFlow Lite	■ Reading	35 min · 75 XP
32	Distributed Training and Data Parallelism	■ Reading	35 min · 75 XP
33	Custom Training Loops with GradientTape	■ Reading	35 min · 75 XP
34	Debugging and Profiling TensorFlow Models	■ Reading	35 min · 75 XP
35	Capstone Project: End-to-End NLP Pipeline	■ Project	60 min · 150 XP

Hugging Face Transformers

■ Complete

Slug: **hugging-face-transformers** · Difficulty: **Intermediate** · Duration: **45h planned · 22h lessons** · Lessons: **35** (■29 ■4 ■0 ■2) · Tags: BERT, GPT, Fine-tuning, Tokenizers, HF Hub

Module 1			
01	Introduction to Transformers Architecture	■ Reading	30 min · 75 XP
02	Attention Mechanism Fundamentals	■ Reading	35 min · 75 XP
03	Self-Attention and Multi-Head Attention	■ Reading	35 min · 75 XP
04	Encoder-Decoder Architecture Overview	■ Reading	35 min · 75 XP
05	Understanding BERT and Its Variants	■ Reading	35 min · 75 XP
Module 2			
06	GPT Models and Autoregressive Language Modeling	■ Reading	35 min · 75 XP
07	Tokenization Strategies and Subword Tokenization	■ Reading	35 min · 75 XP
08	Positional Encoding and Embeddings	■ Reading	35 min · 75 XP
09	Transfer Learning with Pretrained Models	■ Reading	35 min · 75 XP
10	Foundation Concepts Review and Practice	■ Exercise	50 min · 100 XP
Module 3			
11	Setting Up Hugging Face Environment and Installation	■ Reading	20 min · 50 XP
12	Loading and Using Pretrained Models	■ Reading	30 min · 75 XP
13	Working with Tokenizers from the Hub	■ Reading	30 min · 75 XP
14	Fine-tuning BERT for Text Classification	■ Reading	35 min · 75 XP
15	Named Entity Recognition with Transformers	■ Reading	35 min · 75 XP
Module 4			
16	Question Answering with SQuAD Dataset	■ Reading	35 min · 75 XP
17	Semantic Similarity and Sentence Embeddings	■ Reading	35 min · 75 XP
18	Core Skills Checkpoint Exercise	■ Exercise	50 min · 100 XP
19	Sequence-to-Sequence Models for Translation	■ Reading	35 min · 75 XP
20	Building a Complete NLP Pipeline Project	■ Project	60 min · 150 XP
Module 5			
21	Custom Training Loops with PyTorch	■ Reading	35 min · 75 XP
22	Distributed Training and Mixed Precision	■ Reading	35 min · 75 XP
23	Optimizing Model Performance and Inference Speed	■ Reading	35 min · 75 XP
24	Model Quantization and Pruning Techniques	■ Reading	35 min · 75 XP

25	Advanced Skills Assessment	■ Exercise	50 min · 100 XP
Module 6			
26	Zero-Shot and Few-Shot Learning Strategies	■ Reading	35 min · 75 XP
27	Prompt Engineering and In-Context Learning	■ Reading	35 min · 75 XP
28	Multi-Task Learning with Transformers	■ Reading	35 min · 75 XP
29	Domain Adaptation and Transfer Learning Advanced	■ Reading	35 min · 75 XP
30	Advanced Optimization Techniques	■ Exercise	50 min · 100 XP
Module 7			
31	Vision Transformers and Multimodal Models	■ Reading	35 min · 75 XP
32	Efficient Transformers and Long-Context Modeling	■ Reading	35 min · 75 XP
33	Deploying Models to Production with HF Hub	■ Reading	35 min · 75 XP
34	Monitoring and Evaluating Model Performance	■ Reading	35 min · 75 XP
35	Capstone: Advanced Transformer Application Project	■ Project	60 min · 150 XP

Large Language Models

■ Complete

Slug: [large-language-models](#) · Difficulty: **Advanced** · Duration: **60h planned** · **22h lessons** · Lessons: **35** (■29 ■4 ■0 ■2) · Tags: LLMs, Transformers, RLHF, RAG, Deployment

Module 1			
01	Introduction to Large Language Models	■ Reading	30 min · 75 XP
02	History and Evolution of Neural Language Models	■ Reading	30 min · 75 XP
03	Fundamentals of Natural Language Processing	■ Reading	30 min · 75 XP
04	Understanding Transformer Architecture	■ Reading	35 min · 75 XP
05	Attention Mechanisms Explained	■ Reading	35 min · 75 XP
Module 2			
06	Word Embeddings and Vector Representations	■ Reading	35 min · 75 XP
07	Tokenization and Preprocessing Techniques	■ Reading	30 min · 75 XP
08	Training Data and Dataset Considerations	■ Reading	35 min · 75 XP
09	Loss Functions and Optimization in LLMs	■ Reading	35 min · 75 XP
10	Foundation Concepts Assessment	■ Exercise	50 min · 100 XP
Module 3			
11	Transfer Learning and Pre-training Strategies	■ Reading	35 min · 75 XP
12	Fine-tuning Language Models for Specific Tasks	■ Reading	35 min · 75 XP
13	Prompt Engineering Fundamentals	■ Reading	30 min · 75 XP
14	Few-shot and Zero-shot Learning	■ Reading	35 min · 75 XP
15	Chain of Thought Prompting Techniques	■ Reading	35 min · 75 XP
Module 4			
16	Model Evaluation Metrics and Benchmarks	■ Reading	35 min · 75 XP
17	Scaling Laws and Model Size Trade-offs	■ Reading	35 min · 75 XP
18	Core Skills Practice Exercise	■ Exercise	50 min · 100 XP
19	Inference Optimization and Quantization	■ Reading	35 min · 75 XP
20	Building LLM-powered Applications Project	■ Project	60 min · 150 XP
Module 5			
21	API Integration and Deployment	■ Reading	35 min · 75 XP

22	Safety, Bias, and Ethical Considerations	■ Reading	35 min · 75 XP
23	Hallucinations and Reliability Issues	■ Reading	35 min · 75 XP
24	Retrieval-Augmented Generation (RAG)	■ Reading	35 min · 75 XP
25	Advanced Skills Consolidation Exercise	■ Exercise	50 min · 100 XP
Module 6			
26	Multi-modal Language Models	■ Reading	35 min · 75 XP
27	Parameter-Efficient Fine-tuning Methods	■ Reading	35 min · 75 XP
28	Mixture of Experts and Sparse Models	■ Reading	35 min · 75 XP
29	Alignment and Reinforcement Learning from Human Feedback	■ Reading	35 min · 75 XP
30	Advanced Topics Application Exercise	■ Exercise	50 min · 100 XP
Module 7			
31	Emerging Architectures and Future Directions	■ Reading	35 min · 75 XP
32	Efficient Inference at Scale	■ Reading	35 min · 75 XP
33	LLM Interpretability and Analysis	■ Reading	35 min · 75 XP
34	Building Production-Ready LLM Systems	■ Reading	35 min · 75 XP
35	Capstone: Advanced LLM Development Project	■ Project	60 min · 150 XP

Retrieval-Augmented Generation

■ Complete

Slug: **retrieval-augmented-generation** · Difficulty: **Advanced** · Duration: **40h planned** · **22h lessons** · Lessons: **35** (■29 ■4 ■0 ■2) · Tags: RAG, Embeddings, Vector DBs, LangChain, RAGAS

Module 1			
01	Introduction to RAG and Knowledge Retrieval	■ Reading	30 min · 75 XP
02	Embeddings and Vector Representations	■ Reading	35 min · 75 XP
03	Vector Databases: Pinecone, Weaviate, and Chroma	■ Reading	35 min · 75 XP
04	Document Chunking Strategies	■ Reading	35 min · 75 XP
05	Semantic Search and Similarity Metrics	■ Reading	35 min · 75 XP
Module 2			
06	Building a Basic RAG Pipeline	■ Reading	35 min · 75 XP
07	Retrieval Evaluation: Precision, Recall, and NDCG	■ Reading	35 min · 75 XP
08	Hybrid Search: BM25 and Dense Retrieval	■ Reading	35 min · 75 XP
09	Multi-Stage Retrieval and Reranking	■ Reading	35 min · 75 XP
10	Core RAG Skills Exercise	■ Exercise	50 min · 100 XP
Module 3			
11	Advanced Chunking: Semantic and Hierarchical	■ Reading	35 min · 75 XP
12	Query Expansion and Reformulation Techniques	■ Reading	35 min · 75 XP
13	Hypothetical Document Embeddings (HyDE)	■ Reading	35 min · 75 XP
14	Contextual Compression and Filtering	■ Reading	35 min · 75 XP
15	Multi-Hop and Iterative Retrieval	■ Reading	35 min · 75 XP
Module 4			
16	RAG Evaluation with the RAGAS Framework	■ Reading	35 min · 75 XP
17	Handling Conflicting and Contradictory Documents	■ Reading	35 min · 75 XP
18	Intermediate RAG Checkpoint	■ Exercise	50 min · 100 XP
19	GraphRAG: Knowledge Graph Integration	■ Reading	35 min · 75 XP

20	End-to-End Project: Document Q&A; System	■ Project	60 min · 150 XP
Module 5			
21	Streaming RAG Responses	■ Reading	35 min · 75 XP
22	Long-Context RAG with Late Chunking	■ Reading	35 min · 75 XP
23	Domain-Specific RAG Fine-Tuning	■ Reading	35 min · 75 XP
24	Cache-Augmented Generation	■ Reading	35 min · 75 XP
25	Advanced RAG Patterns Exercise	■ Exercise	50 min · 100 XP
Module 6			
26	Production RAG Architecture and Scalability	■ Reading	35 min · 75 XP
27	Indexing Large Corpora Efficiently	■ Reading	35 min · 75 XP
28	Self-Corrective RAG (CRAG)	■ Reading	35 min · 75 XP
29	Agentic RAG: Dynamic and Adaptive Retrieval	■ Reading	35 min · 75 XP
30	Advanced Production Exercise	■ Exercise	50 min · 100 XP
Module 7			
31	Multimodal RAG: Images, Tables, and PDFs	■ Reading	35 min · 75 XP
32	Real-Time RAG with Streaming Ingestion	■ Reading	35 min · 75 XP
33	RAG Security and PII Handling	■ Reading	35 min · 75 XP
34	Cost Optimization for RAG Pipelines	■ Reading	35 min · 75 XP
35	Capstone: Enterprise Knowledge Assistant	■ Project	60 min · 150 XP

AI Agents & Agentic Workflows

■ Complete

Slug: [ai-agents-agentic-workflows](#) · Difficulty: **Advanced** · Duration: **50h planned** · **22h lessons** · Lessons: **35** (■29 ■4 ■0 ■2) · Tags: Agents, ReAct, LangGraph, AutoGen, CrewAI

Module 1			
01	Introduction to AI Agents and Agentic Systems	■ Reading	30 min · 75 XP
02	Agent Architectures: Reactive, Deliberative, and Hybrid	■ Reading	35 min · 75 XP
03	Memory and State Management in Agents	■ Reading	35 min · 75 XP
04	Tool Use and Function Calling	■ Reading	35 min · 75 XP
05	Planning and Goal Decomposition	■ Reading	35 min · 75 XP
Module 2			
06	The ReAct Pattern: Reasoning and Acting	■ Reading	35 min · 75 XP
07	Chain of Thought for Agent Reasoning	■ Reading	35 min · 75 XP
08	Multi-Agent Systems Overview	■ Reading	35 min · 75 XP
09	Agent Communication Protocols	■ Reading	35 min · 75 XP
10	Core Concepts Exercise: Build a Simple Agent	■ Exercise	50 min · 100 XP
Module 3			
11	LangChain Agents Deep Dive	■ Reading	35 min · 75 XP
12	LangGraph for Stateful Agents	■ Reading	35 min · 75 XP
13	AutoGen: Multi-Agent Conversations	■ Reading	35 min · 75 XP
14	CrewAI: Role-Based Agent Collaboration	■ Reading	35 min · 75 XP
15	Custom Tool Development for Agents	■ Reading	35 min · 75 XP
Module 4			
16	Context Window Management and Summarization	■ Reading	35 min · 75 XP

17	Agent Orchestration and Delegation Patterns	■ Reading	35 min · 75 XP
18	Intermediate Skills Checkpoint	■ Exercise	50 min · 100 XP
19	Advanced Planning: Tree of Thoughts	■ Reading	35 min · 75 XP
20	End-to-End Project: Research Agent	■ Project	60 min · 150 XP
Module 5			
21	Agent Evaluation and Benchmarking	■ Reading	35 min · 75 XP
22	Error Handling and Agent Recovery Strategies	■ Reading	35 min · 75 XP
23	Human-in-the-Loop Agent Design	■ Reading	35 min · 75 XP
24	Agent Security and Trust Boundaries	■ Reading	35 min · 75 XP
25	Advanced Agent Patterns Exercise	■ Exercise	50 min · 100 XP
Module 6			
26	Production Deployment of AI Agents	■ Reading	35 min · 75 XP
27	Monitoring and Observability for Agents	■ Reading	35 min · 75 XP
28	Cost Optimization for Agent Pipelines	■ Reading	35 min · 75 XP
29	Agent Testing and Quality Assurance	■ Reading	35 min · 75 XP
30	Advanced Deployment Exercise	■ Exercise	50 min · 100 XP
Module 7			
31	Multimodal Agents: Vision and Audio Capabilities	■ Reading	35 min · 75 XP
32	Autonomous Code Generation Agents	■ Reading	35 min · 75 XP
33	Domain-Specific Agent Customization	■ Reading	35 min · 75 XP
34	Enterprise Agent Architecture Patterns	■ Reading	35 min · 75 XP
35	Capstone: Production-Ready Agentic System	■ Project	60 min · 150 XP

Machine Learning Fundamentals

■ Pending

Slug: [ml-fundamentals](#) · Difficulty: **Intermediate** · Duration: **38h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Regression, Classification, Clustering, Scikit-Learn

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Generative AI & LLMs

■ Pending

Slug: [generative-ai](#) · Difficulty: **Intermediate** · Duration: **30h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: GPT, Fine-tuning, Embeddings, Multimodal

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

RAG Systems & Vector Search

Pending

Slug: rag · Difficulty: Intermediate · Duration: 24h planned · 3h lessons · Lessons: 5 (2 1 1 1) · Tags: Embeddings, Vector DB, LangChain, LlamaIndex

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

AI Agents & Tool Use

Pending

Slug: ai-agents · Difficulty: Advanced · Duration: 28h planned · 3h lessons · Lessons: 5 (2 1 1 1) · Tags: ReAct, Function Calling, Multi-Agent, AutoGen

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Advanced Prompt Engineering

Pending

Slug: prompt-engineering · Difficulty: Beginner · Duration: 12h planned · 3h lessons · Lessons: 5 (2 1 1 1) · Tags: Chain-of-Thought, Few-Shot, System Prompts

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

MLOps & Model Deployment

Pending

Slug: mlops · Difficulty: Advanced · Duration: 36h planned · 3h lessons · Lessons: 5 (2 1 1 1) · Tags: MLflow, Kubeflow, Docker, Monitoring, Feature Stores

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Databases

PostgreSQL Advanced

■ Pending

Slug: **postgresql** · Difficulty: **Intermediate** · Duration: **28h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Indexing, Query Optimization, Partitioning, JSONB

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

MongoDB & Atlas

■ Pending

Slug: **mongodb** · Difficulty: **Intermediate** · Duration: **22h planned** · **3h lessons** · Lessons: 5 (■2 ■1 ■1 ■1) · Tags: Aggregation, Indexes, Atlas Search, Transactions

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Software Engineering

Data Structures & Algorithms

■ Pending

Slug: **dsa** · Difficulty: **Intermediate** · Duration: **60h planned** · **3h lessons** · Lessons: **5** (■2 ■1 ■1 ■1) · Tags: Arrays, Trees, Graphs, DP, Sorting, Interview Prep

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

System Design

■ Pending

Slug: **system-design** · Difficulty: **Advanced** · Duration: **40h planned** · **3h lessons** · Lessons: **5** (■2 ■1 ■1 ■1) · Tags: Scalability, CAP Theorem, Load Balancing, Caching

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Mobile Development

React Native

■ Pending

Slug: **react-native** · Difficulty: **Intermediate** · Duration: **36h planned** · **3h lessons** · Lessons: **5** (■2 ■1 ■1 ■1) · Tags: Expo, Navigation, Animations, Native Modules

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Flutter & Dart

■ Pending

Slug: **flutter** · Difficulty: **Intermediate** · Duration: **40h planned** · **3h lessons** · Lessons: **5** (■2 ■1 ■1 ■1) · Tags: Widgets, State Management, BLoC, Animations, Dart

Module 1

01	Introduction & Core Concepts	■ Reading	20 min · 50 XP
02	Deep Dive & Advanced Patterns	■ Reading	35 min · 75 XP
03	Hands-on Exercise	■ Exercise	50 min · 100 XP
04	Quiz: Test Your Knowledge	■ Quiz	20 min · 50 XP
05	Capstone Mini Project	■ Project	60 min · 150 XP

Summary by Category

Category	Courses	Complete	Total Lessons	Est. Hours
■ Programming Languages	6	0/6	30	204h
■ Frontend Development	5	0/5	25	152h
■ Backend Development	5	0/5	25	174h
■ Cloud Computing	1	0/1	5	50h
■ DevOps & Infrastructure	3	0/3	15	94h
■ AI & Machine Learning	13	7/13	275	513h
■ Databases	2	0/2	10	50h
■ Software Engineering	2	0/2	10	100h
■ Mobile Development	2	0/2	10	76h
TOTAL	39	7/39	405	1413h